REPORT - ASSIGNMENT 4 - APL

Keyboard Analysis Programming Assignment

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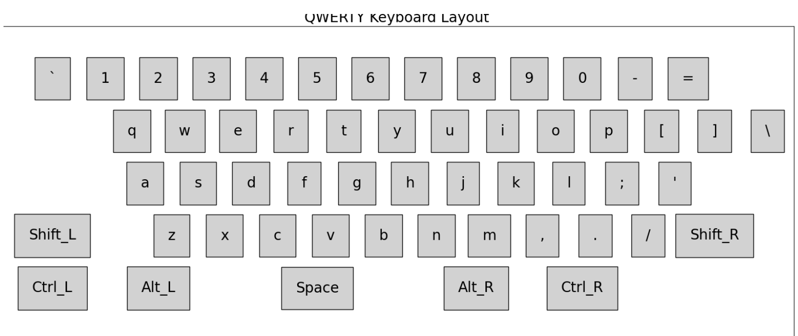
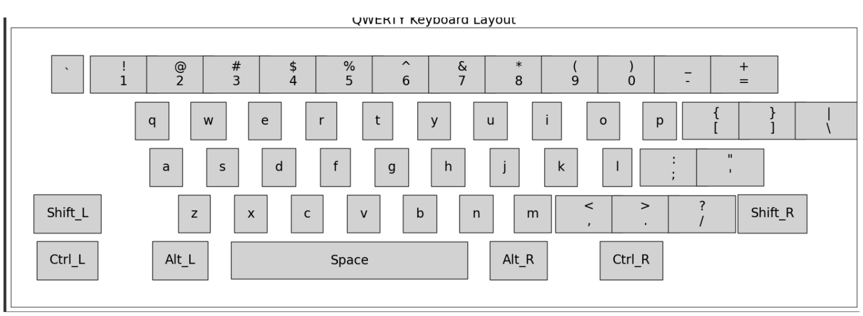
In my approach, I first define the keyboard layout using the keys and characters dictionary and extract the keys of the keyboard ,which should be displayed along with corresponding shift characters or e.g: 1 and !, into a dictionary called `Shift_characters_simplified.`
I do so to make plotting easier.

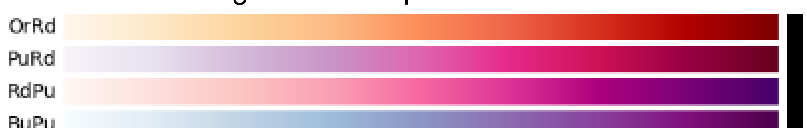
Coming to plotting the layout,





I tried the simple layout as per the keys and characters dictionaries, but didn’t find them visually appealing hence, I added extra keys such as TAB, BACKSPACE, ENTER, this helped adjust the alignment and uniformity in keys. To further enhance the clarity in the image, I did trial and error with many parameters like the width, shape of each key and moved some redundant keys around, so that it would look better and not disturb the distance calculation and got this as my final layout -

Now, coming to the heat map generation. I tried using different color maps and found PuRd o be the best in indicating the heatmap.



The low frequency keys are white to violet and high frequency ones are red. Stored the heatmap data or the frequencies (also called heat) as a 2D matrix to later on help with the color scale plotting.

Later on I captured all this key by key using

```
FuncAnimation(fig, generate_animated_heatmap, frames=len(text_input)
+ 1, interval=500, repeat=False)
```

```
from matplotlib.animation import FuncAnimation
```

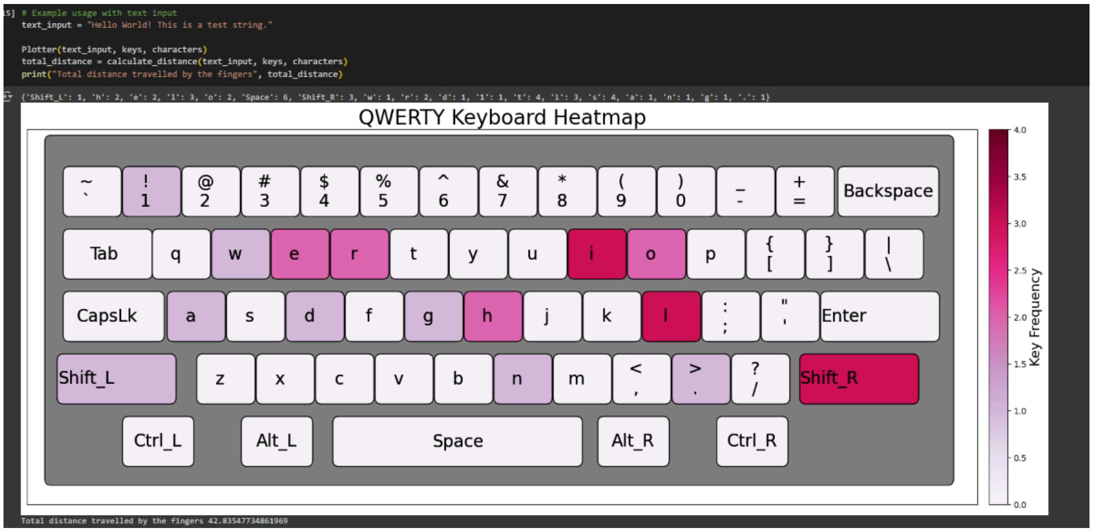
Coming to distance calculation, I calculated total distance by summing up distances for each character in the string. By distances for each character in the string I mean, total length traveled from corresponding start positions/ home row keys to that character key or to those character combination of keys.

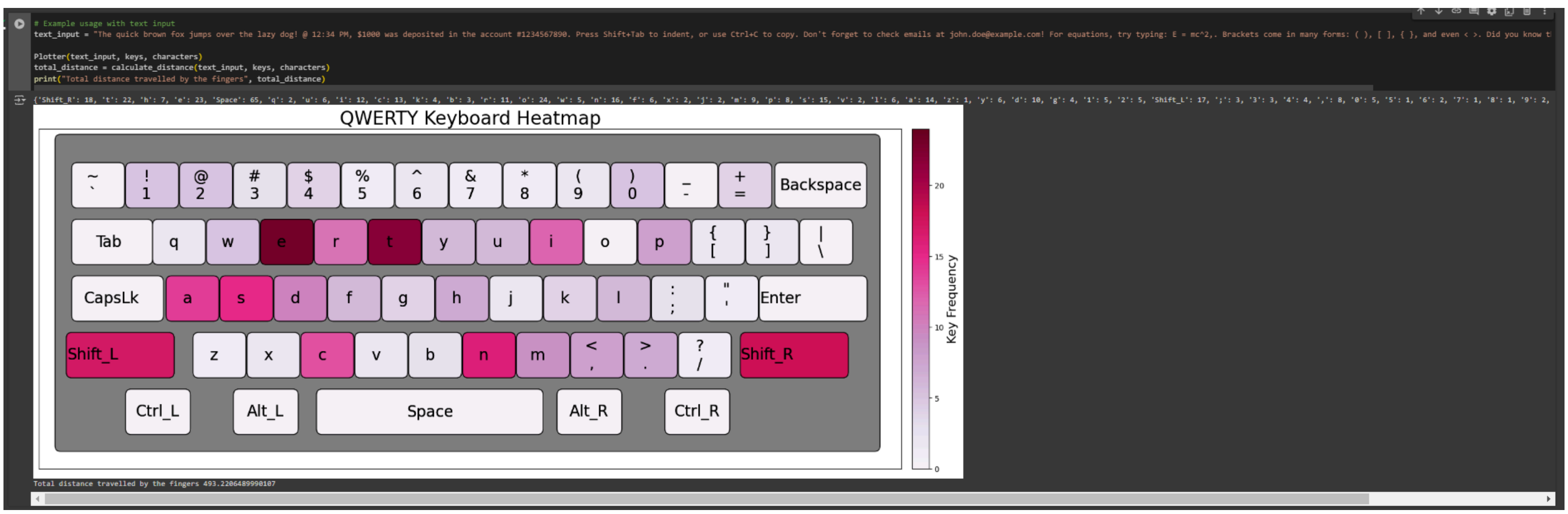
I assumed the space key to have f as the home row and not zero distance from the thumbs.

I also did not include space keys in the heatmap generation to prevent skew in paragraphs as that would tend to be the most used key.

RESULTS :

I have attached the animations and heatmaps for each input I tried -





My colab notebook, incase any issue arises with the images and animations.

EE23B188- Nishitha.D - Assignment 4 - EE2703